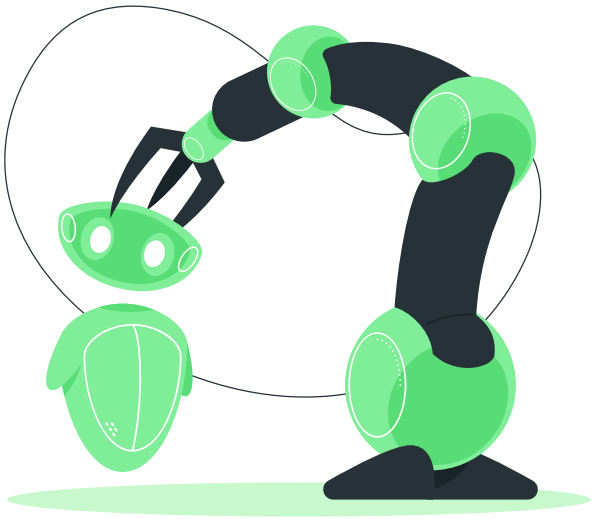
Robotics
CS 3015

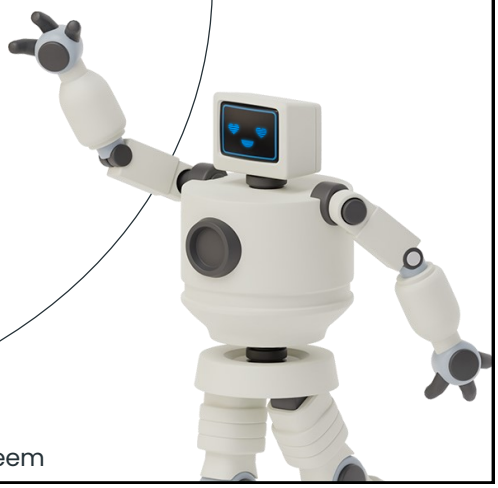
Umarah Qaseem



1

7a

- Practice Numerical problems
- Frame assignment
- Inverse Kinematics
- Derivation of q1 through Geometric Approach



2

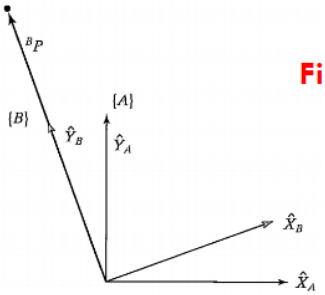
Some Extra Practice Problems



4

Example 1

- Frame {B} that is rotated relative to frame {A} about Z by 30 degrees.



Find ${}^{A}P$?

$$ {}^{B}P = \begin{bmatrix} 0.0 \\ 2.0 \\ 0.0 \end{bmatrix} $$

$$ {}^{A}_{B}R = \begin{bmatrix} 0.866 & -0.500 & 0.000 \\ 0.500 & 0.866 & 0.000 \\ 0.000 & 0.000 & 1.000 \end{bmatrix} $$

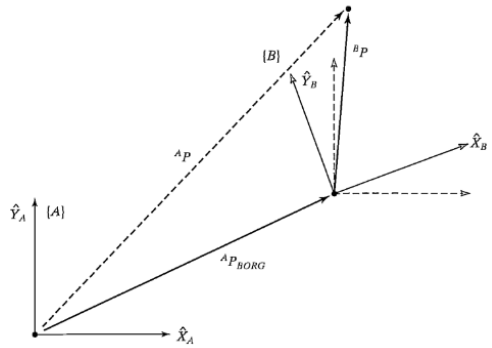
$$ {}^{A}P = {}^{A}_{B}R\,{}^{B}P = \begin{bmatrix} -1.000 \\ 1.732 \\ 0.000 \end{bmatrix} $$

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Example 2

$$ {}^{A}_{B}T = \begin{bmatrix} 0.866 & -0.500 & 0.000 & 10.0 \\ 0.500 & 0.866 & 0.000 & 5.0 \\ 0.000 & 0.000 & 1.000 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

- Figure shows a frame {B}, which is rotated relative to frame {A} about Z by 30 degrees, translated 10 units in X_A, and translated 5 units in Y_A Find ^{A}p, where $^{B}p = [3\ 7\ 0]^T$

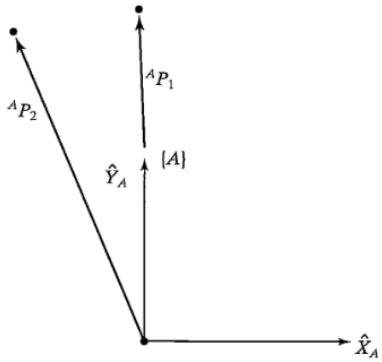


$$ {}^{B}P = \begin{bmatrix} 3.0 \\ 7.0 \\ 0.0 \end{bmatrix} $$

$$ {}^{A}P = {}^{A}_{B}T\ {}^{B}P = \begin{bmatrix} 9.098 \\ 12.562 \\ 0.000 \end{bmatrix} $$

Example 3

- Figure shows a vector $^{A}P_1=[0\ 2\ 0]$. We wish to compute the vector obtained by rotating this vector about Z by 30 degrees. Call the new vector $^{A}P_2$

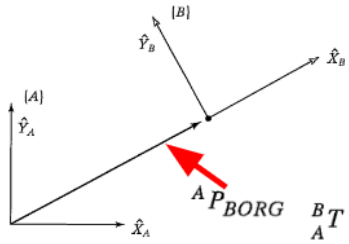


$$ R_z(30.0) = \begin{bmatrix} 0.866 & -0.500 & 0.000 \\ 0.500 & 0.866 & 0.000 \\ 0.000 & 0.000 & 1.000 \end{bmatrix} $$

$$ {}^{A}P_2 = R_z(30.0)\ {}^{A}P_1 = \begin{bmatrix} -1.000 \\ 1.732 \\ 0.000 \end{bmatrix} $$

Example 4

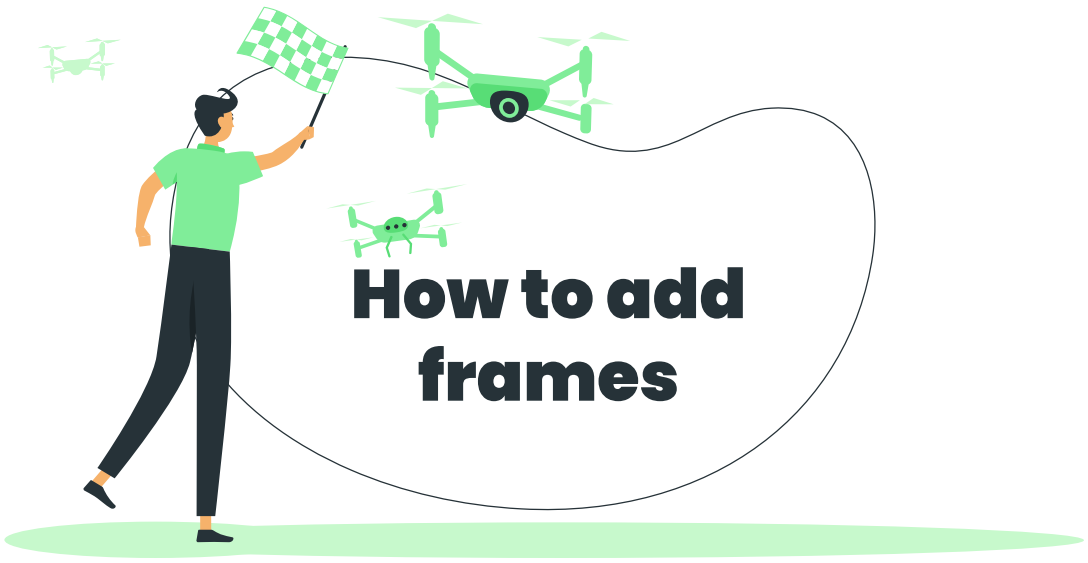
- Figure shows a frame {B} that is rotated relative to frame {A} about Z by 30 degrees and translated four units in X_A and three units in Y_A . Find ${}^{A}_{B}T$ and ${}^{B}_{A}T$



$$ {}^{A}_{B}T = \begin{bmatrix} 0.866 & -0.500 & 0.000 & 4.0 \\ 0.500 & 0.866 & 0.000 & 3.0 \\ 0.000 & 0.000 & 1.000 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{B}_{A}T = \left[\begin{array}{ccc|c} & {}^{A}_{B}R^{T} & & -{}^{A}_{B}R^{T}\,{}^{A}P_{BORG} \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \quad {}^{B}_{A}T = \begin{bmatrix} 0.866 & 0.500 & 0.000 & -4.964 \\ -0.500 & 0.866 & 0.000 & -0.598 \\ 0.000 & 0.000 & 1.000 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{B}_{A}T = {}^{A}_{B}T^{-1} $$



How to add frames

Four Rules

- The z-axis is the axis of rotation for a revolute joint.
- The x-axis must be perpendicular to both the current z-axis and the previous z-axis.
- The y-axis is determined from the x-axis and z-axis by using the right-hand coordinate system.
- The x-axis must intersect the previous z-axis (rule does not apply to frame 0).

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Examples for frames



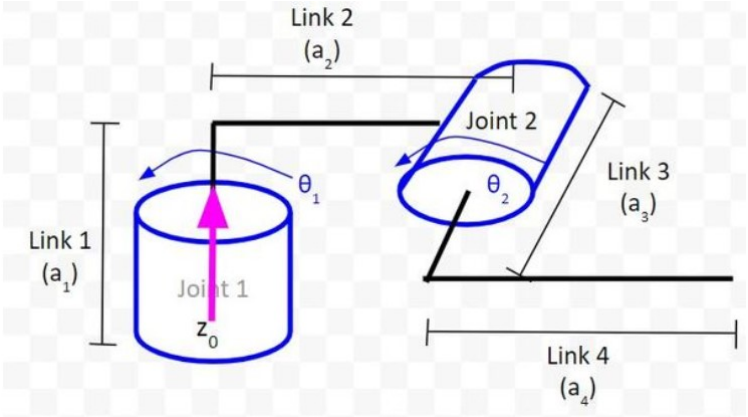
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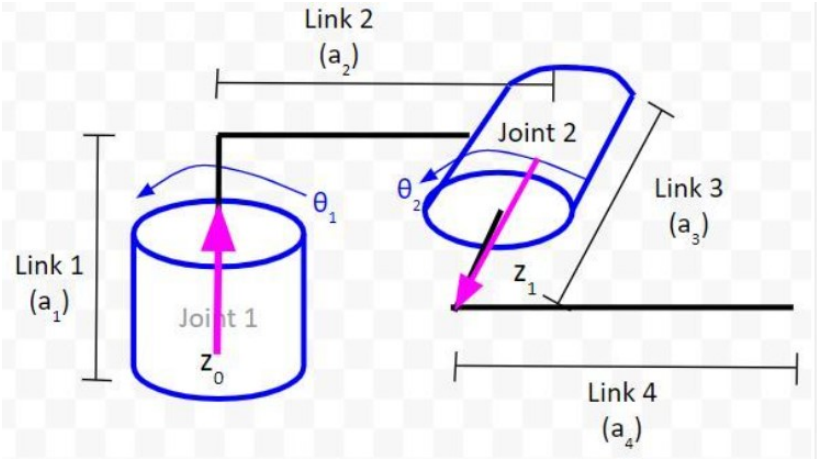
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Let's take an example

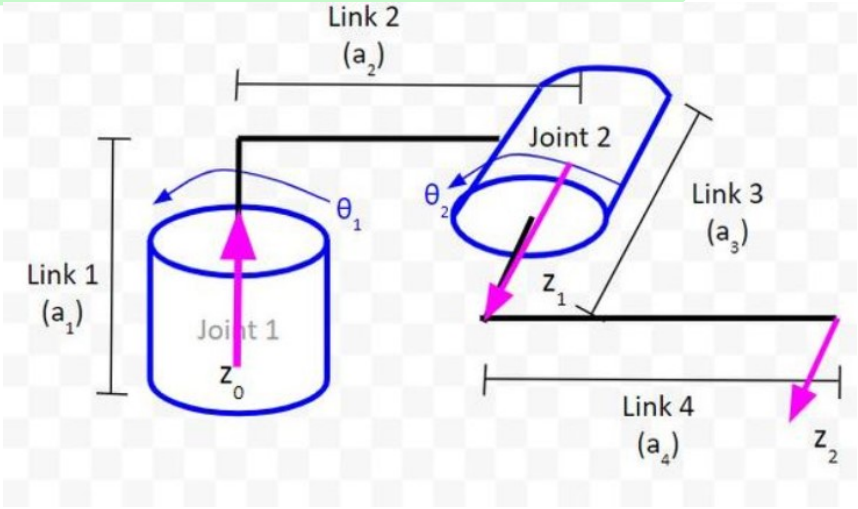
- The z-axis is the axis of rotation for a revolute joint



Draw z1



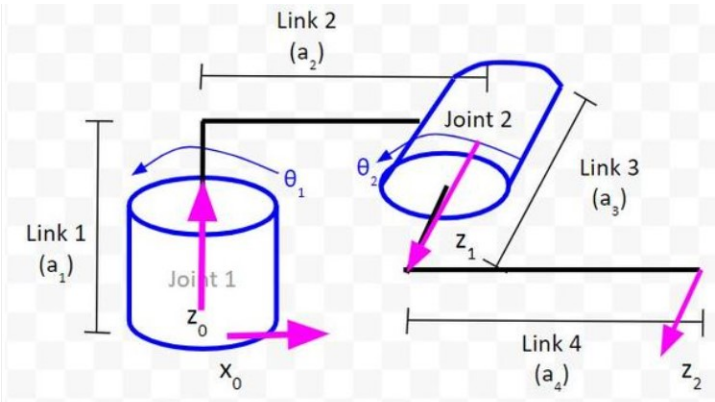
Draw z2



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x0

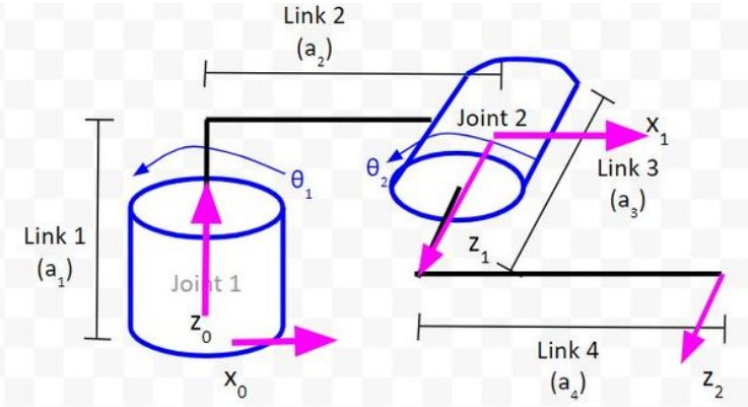
- 2- The x-axis must be perpendicular to both the current z-axis and the previous z-axis.



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x1

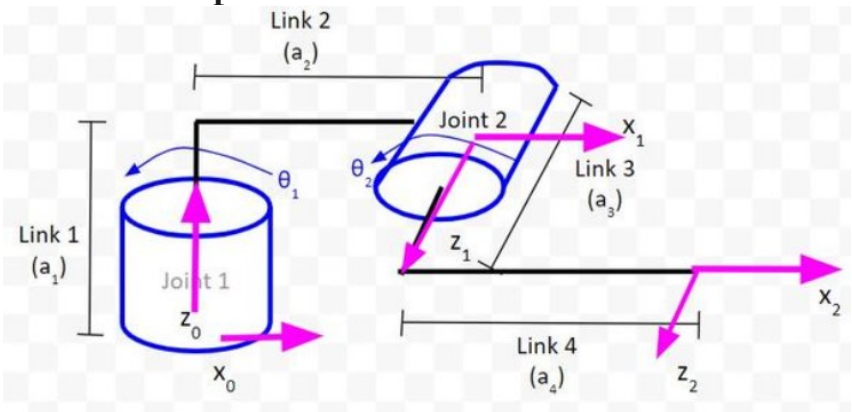
- 2- The x-axis must be perpendicular to both the current z-axis and the previous z-axis.



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x2

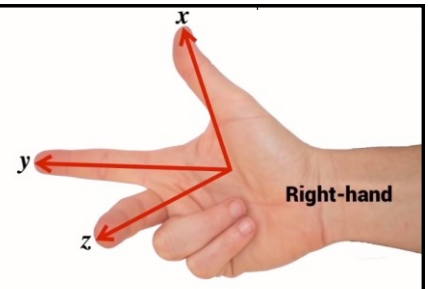
- 2- The x-axis must be perpendicular to both the current z-axis and the previous z-axis.

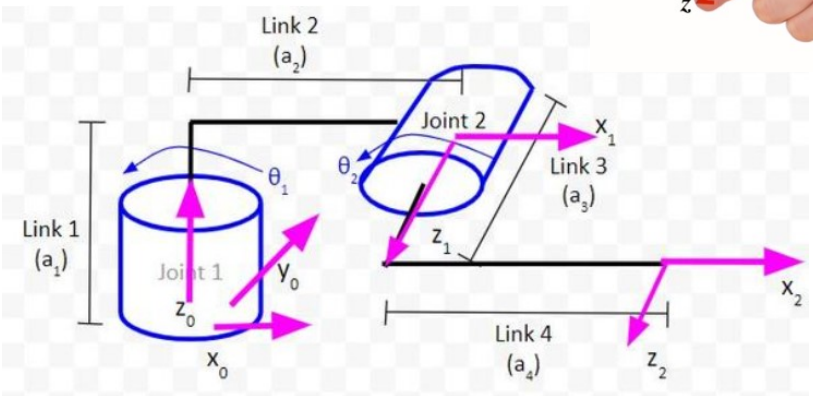


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y0

- 3- Use right hand rule for y



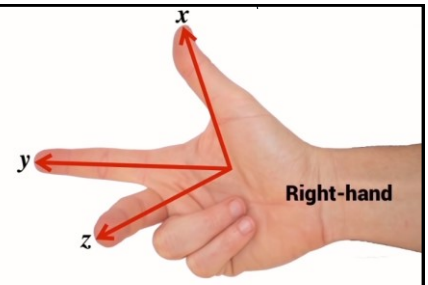


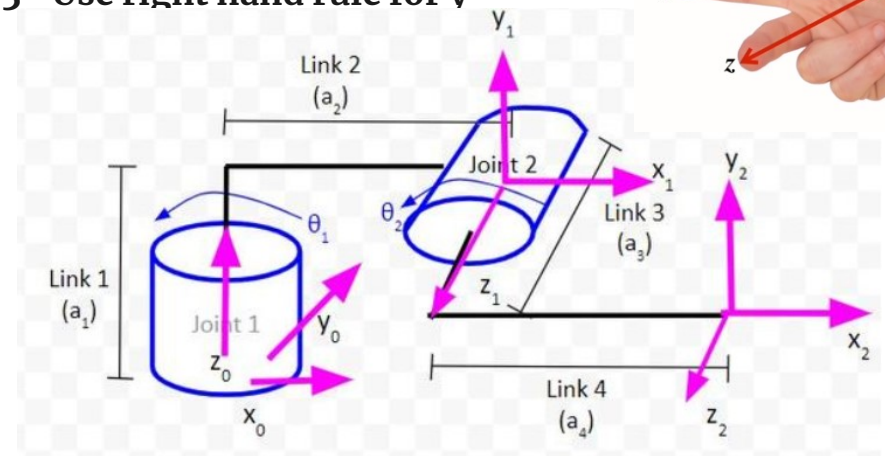
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Y1 & y2

- 3- Use right hand rule for y



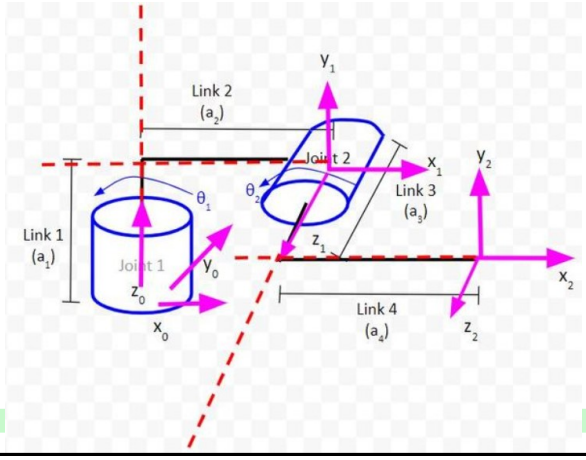


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Last Rule

- The x-axis must intersect the previous z-axis (rule does not apply to frame 0).

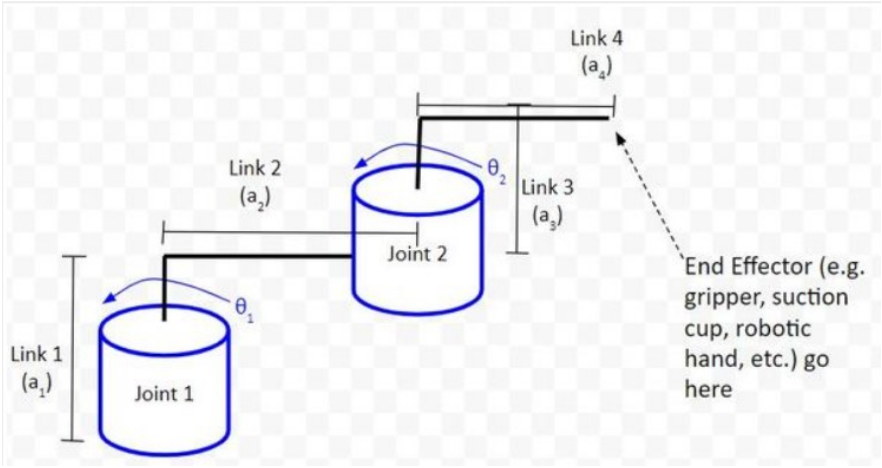


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Example 2

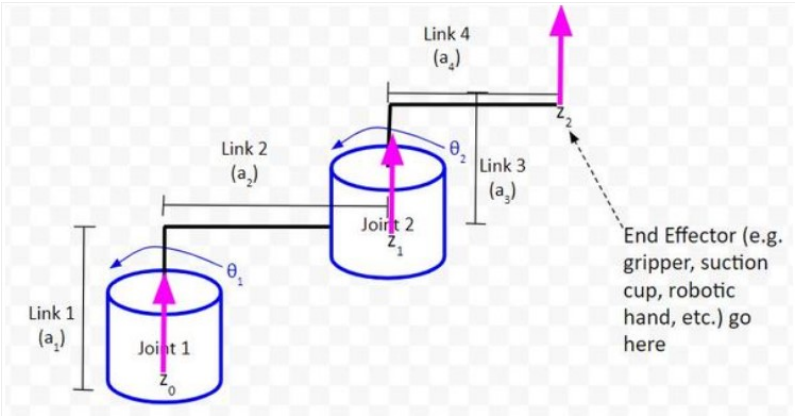
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Example 2



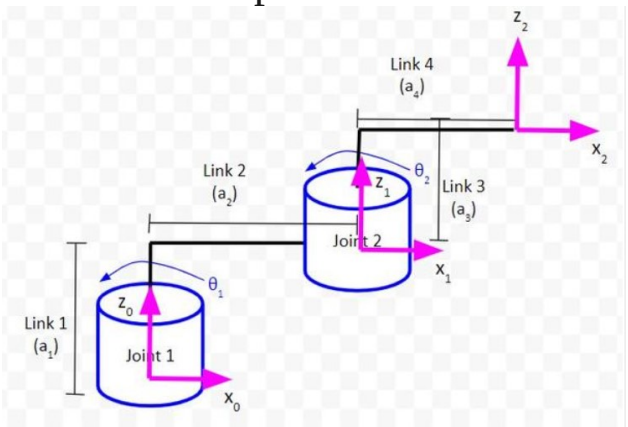
Example 2

- Rule #1: The z-axis is the axis of rotation for a revolute joint



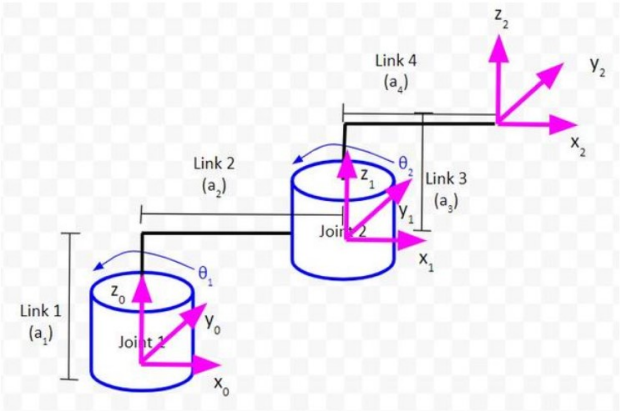
Example 2

- Rule #2: The x-axis must be perpendicular to both the current z-axis and the previous z-axis.



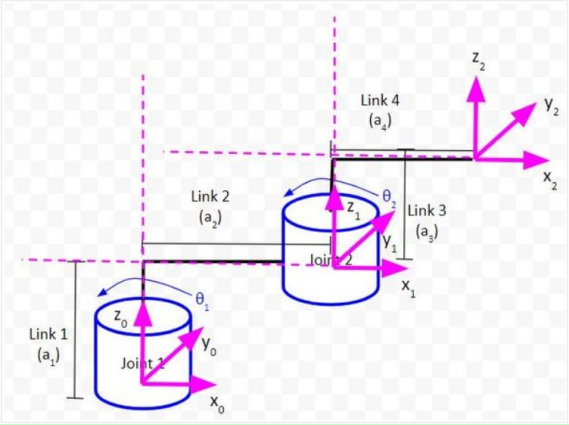
Example 2

- Rule #3: The y-axis is determined from the x-axis and z-axis by using the right-hand coordinate system.



Example 2

- Rule #4: The x-axis must intersect the previous z-axis (rule does not apply to frame 0).

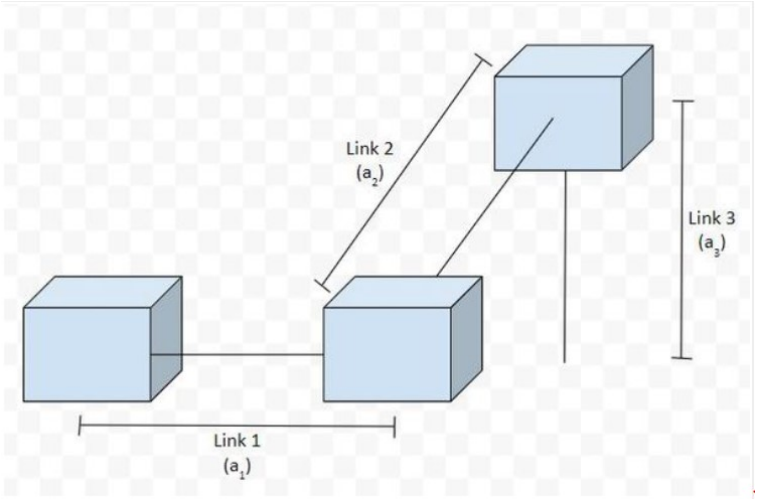


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Example 3

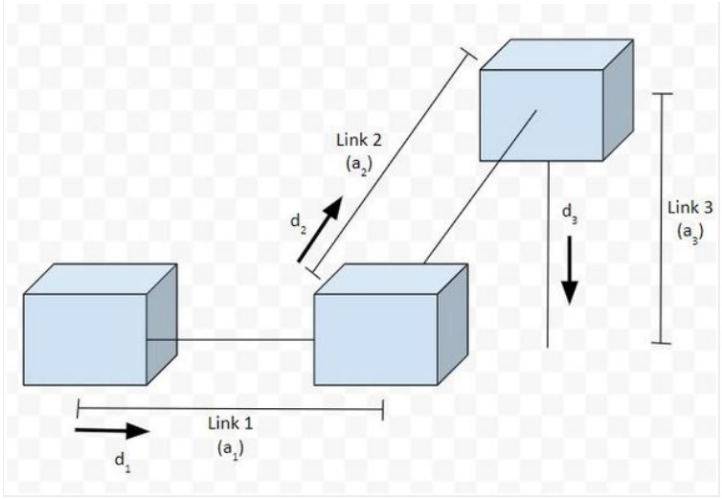
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Example 3



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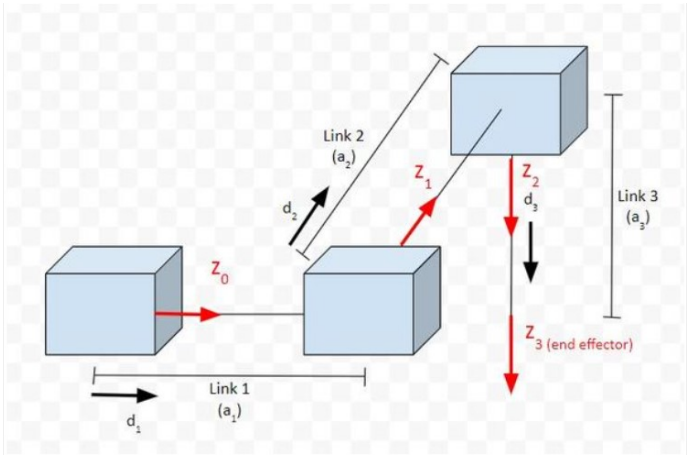
Example 3



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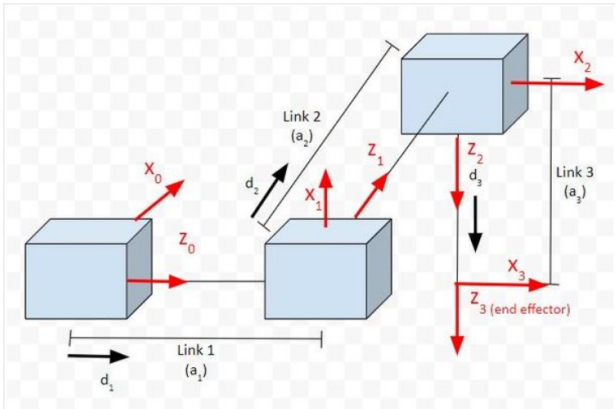
Example 3

- Rule #1: For a prismatic joint, the z-axis has to be the direction of motion



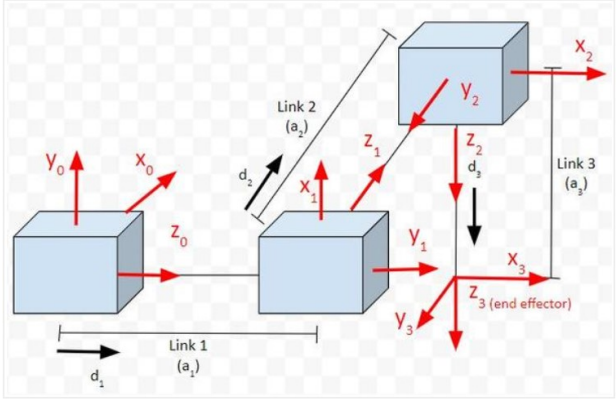
Example 3

- Rule #2: The x-axis must be perpendicular to both the current z-axis and the previous z-axis.



Example 3

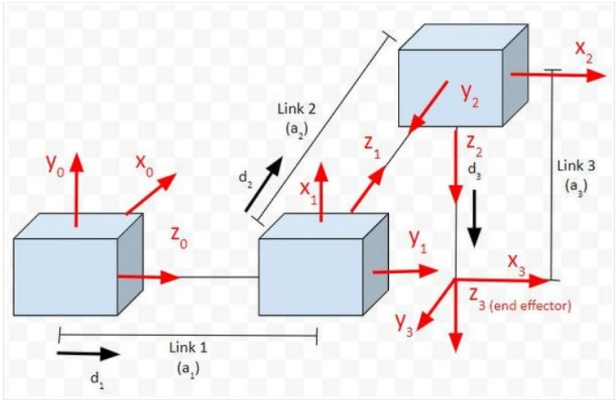
- Rule #3: The y-axis is determined from the x-axis and z-axis by using the right-hand coordinate system.



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Example 3

- Rule #4: The x-axis must intersect the previous z-axis (rule does not apply to frame 0).

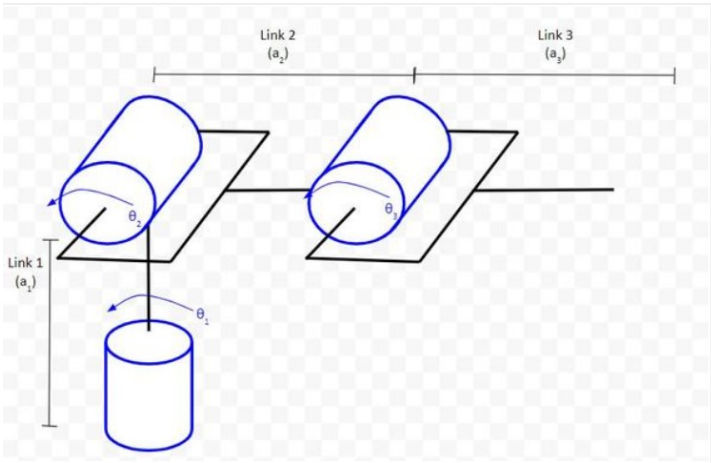


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Home Task

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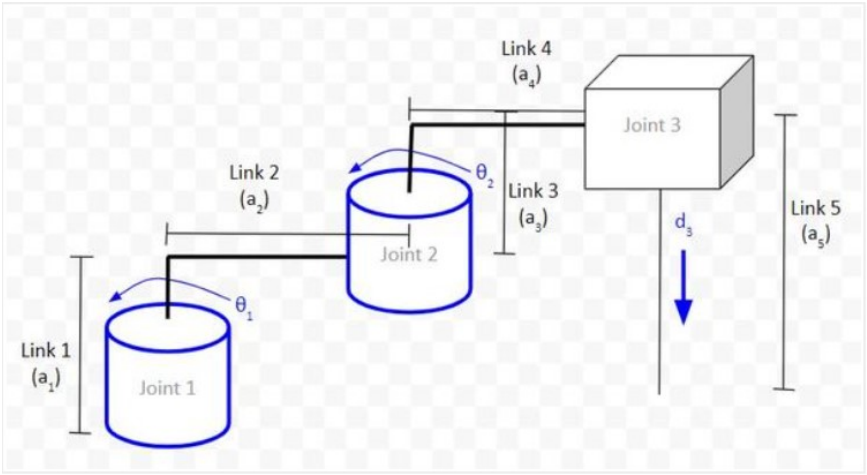
Example 4 – Task 1



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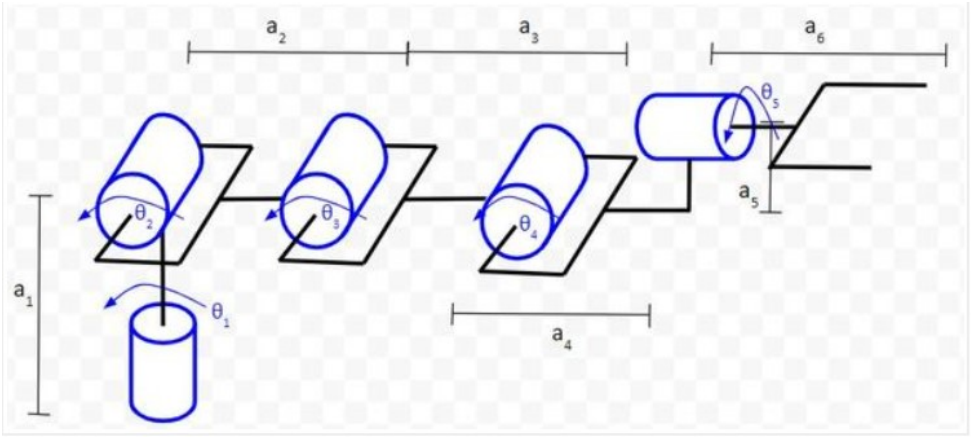
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Example 5 – Task 2



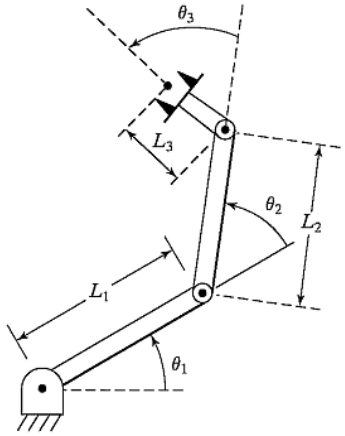
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Example 6 – Task 3



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Example 7 – Task 4



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Fill DH table for task 4 as well

theta	d	a	twist

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Home Task

Submit Task 1 to Task 4 (handwritten) in next class with NEAT diagrams

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Inverse Kinematics

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Inverse Kinematics

- What joint angles to set to achieve a certain end-effector pose.

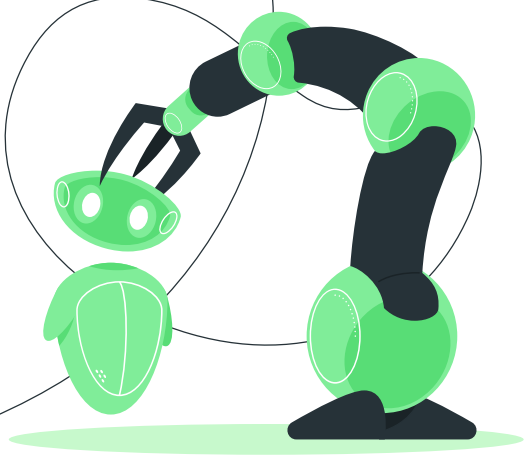
$$\xi_N = \mathcal{K}(\boldsymbol{q})$$

$$\boldsymbol{q} = \{q_j, j \in [1 \cdots N]\}$$

$$\boldsymbol{q} = \mathcal{K}^{-1}(\xi_N)$$

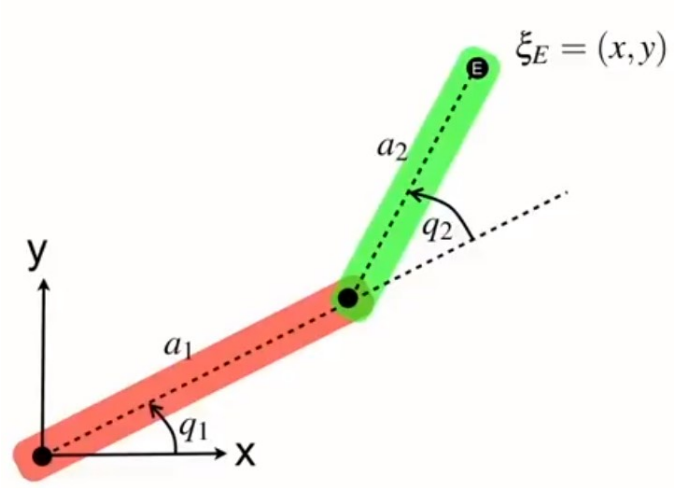
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Inverse Kinematics Geometric Approach



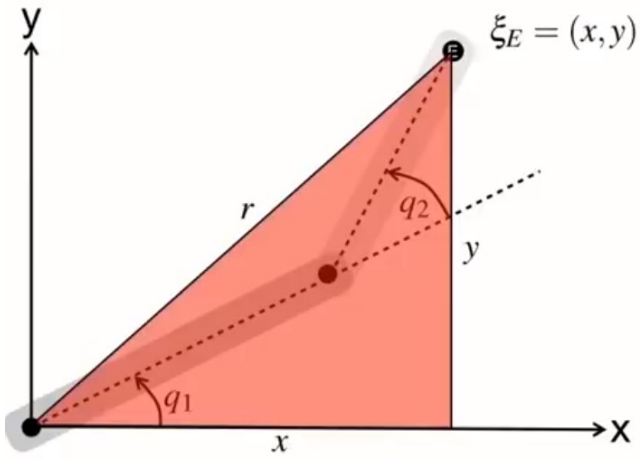
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IK for 2 Joint Arm – Geometric Approach



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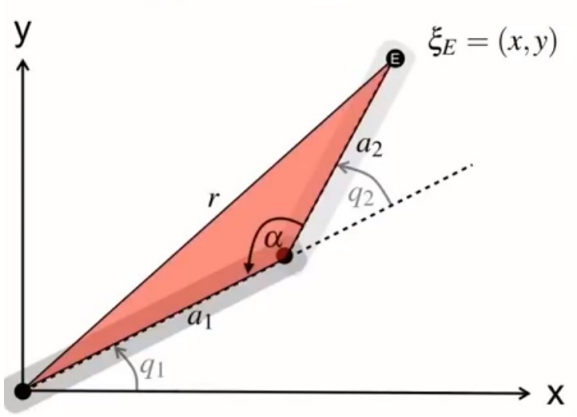
IK or 2 joint arm – Geometric Approach



$$r^2 = x^2 + y^2$$

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IK or 2 joint arm – Geometric Approach



***We need to find angle alpha so we apply cosine rule**

Cosine Rule

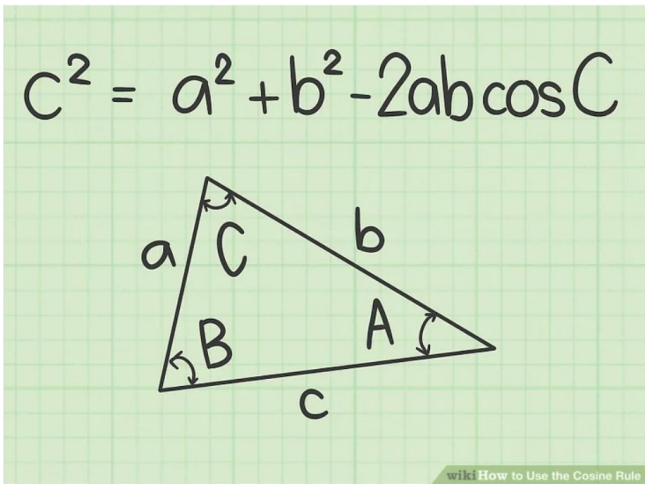
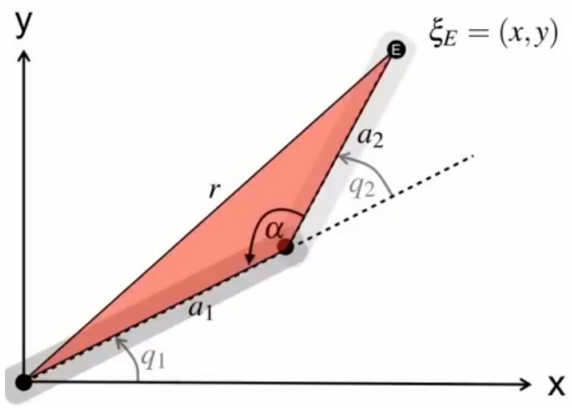


Image Credits: wikiHow

IK or 2 joint arm – Geometric Approach



$$r^2 = x^2 + y^2$$

$$r^2 = a_1^2 + a_2^2 - 2a_1 a_2 \cos\alpha$$

$$\cos\alpha = \frac{a_1^2 + a_2^2 - r^2}{2a_1 a_2} = \frac{a_1^2 + a_2^2 - x^2 - y^2}{2a_1 a_2}$$

$$q_2 = \pi - \alpha$$

$$\cos q_2 = -\cos\alpha$$

$$\cos q_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1 a_2}$$

Credits

- This template is customized from a Slidesgo template.
- This lecture is made from the book “RVC by Professor Peter Corke”
- Some slides are inspired by Professor Peter Corke’s work.
- Rest of the material is self made and compiled from internet resources.

slidesgo